

$$x(x \cos x + 2 \sin x)$$

$$f''(x) =$$

$$2x \cos x + x^2(-\sin x)$$

$$+ 2 \sin x + 2x \cos x$$

$$= 4x \cos x - x^2 \sin x + 2 \sin x$$

Set 5

2024 slot 3

1) Mode when frequency interval is given

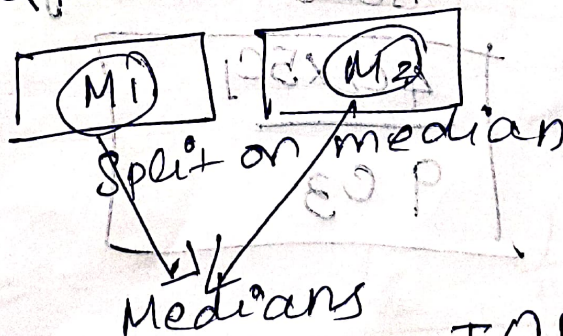
$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

frequency band's lowest limit

highest
preceeding
succeeding

2) see ML Note

IQR



$$M_2 - M_1 = \text{IQR}$$

3) odd fn. n - no sign change
 $f(-x) = -f(x)$

4 confusion matrix
repeated.

sensitivity - Recall

$$\text{specificity} \Rightarrow \frac{TN}{TN + FP}$$

set 6

2025

FN